

Literature Status: The Lieb–Solovej Bergman Conjecture

Automatic Math Discovery Run `fa_banach_001`

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Status

This is a **literature_already_answered** packet for the full problem. It records a separate later paper that explicitly resolves the Lieb–Solovej conjecture studied in arXiv:2010.14809. It is not a new proof packet.

Source Conjecture

The source is D. Békollè, J. Gonessa, and B. F. Sehba, *About a conjecture of Lieb–Solovej*, arXiv:2010.14809 [1]. Equation (1.1), beginning on PDF page 1 and continuing on PDF page 2, asks whether for every real $s \geq 1$ and every $F \in A^2(\Pi^+)$,

$$\int_{\Pi^+} |F(x + iy)|^{2s} y^{2s-2} dx dy \leq \frac{\pi^{1-s}}{(2s-1)2^{2s-2}} \left(\int_{\Pi^+} |F(x + iy)|^2 dx dy \right)^s. \quad (1)$$

The text on PDF page 2 explains that the open part is the non-integer case and says that the authors do not know a general proof. Section 4 gives the equivalent disk form

$$\int_{\mathbb{D}} |f(z)|^{2s} (1 - |z|^2)^{2s-2} dx dy \leq \frac{\pi^{1-s}}{2s-1} \left(\int_{\mathbb{D}} |f(z)|^2 dx dy \right)^s. \quad (2)$$

Supporting Answer

The supporting paper is A. Kulikov, *Functionals with extrema at reproducing kernels*, arXiv:2203.12349, published in *Geometric and Functional Analysis* 32 (2022), 938–949 [2]. Its abstract and introduction explicitly state that it confirms the Lieb–Solovej Wehrl-type/Bergman contractivity conjecture.

Kulikov normalizes hyperbolic measure and weighted Bergman norms by

$$dm(z) = \frac{dx dy}{\pi(1 - |z|^2)^2}, \quad \|f\|_{A_\alpha^p}^p = \int_{\mathbb{D}} (\alpha - 1) |f(z)|^p (1 - |z|^2)^\alpha dm(z).$$

Theorem 1.2 and Corollary 1.3 on supporting-PDF page 3 prove that

$$\|f\|_{A_\beta^q} \leq \|f\|_{A_\alpha^p} \quad \text{when} \quad 0 < p < q, \quad 1 < \alpha < \beta, \quad \frac{p}{\alpha} = \frac{q}{\beta}.$$

Exact Identification

Fix $s > 1$ and substitute

$$(p, \alpha) = (2, 2), \quad (q, \beta) = (2s, 2s).$$

The ratio condition is satisfied, and the definitions give

$$\|f\|_{A_2^2}^2 = \frac{1}{\pi} \int_{\mathbb{D}} |f|^2 dx dy$$

and

$$\|f\|_{A_{2s}^{2s}}^{2s} = \frac{2s-1}{\pi} \int_{\mathbb{D}} |f|^{2s} (1 - |z|^2)^{2s-2} dx dy.$$

Raising Kulikov's contractive inequality to the power $2s$ and rearranging produces exactly (2), including its sharp constant. The Cayley transfer used in the source paper then gives (1). The endpoint $s = 1$ is an identity. Thus arXiv:2203.12349 answers the full conjecture.

Scope and Search Evidence

No part of the stated inequality remains open in this packet's scope. Kulikov's theorem is stronger: it treats general convex functionals and identifies normalized reproducing kernels as maximizers. Local cheap indexes had no exact record for arXiv:2010.14809 at reservation time. A bounded search on August 11, 2026 for the exact conjecture name, the sharp Bergman embedding, and its $SU(1,1)$ formulation found arXiv:2203.12349; its explicit attribution and the normalization calculation above determine the classification as an already-known later answer.

References

- [1] D. Békollè, J. Gonessa, and B. F. Sehba, *About a conjecture of Lieb–Solovej*, arXiv:2010.14809 (2020).
- [2] A. Kulikov, *Functionals with extrema at reproducing kernels*, Geom. Funct. Anal. **32** (2022), 938–949; arXiv:2203.12349; doi:10.1007/s00039-022-00608-5.